

BELFORT FONTAINE, France

WMO: 072950

Lat: 47.633N

Lon: 6.867E

Elev: 1388

StdP: 13.97

Time Zone: 1.00 (EUC)

Period: 99-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.0	21.2	13.1	11.3	20.8	16.0	13.0	23.5	21.0	48.1	18.9	46.8	2.6	80	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	86.2	67.6	82.8	66.8	79.8	66.0	69.7	82.3	68.3	80.3	66.9	77.7	3.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	98.8	75.1	64.1	94.2	73.2	62.8	90.1	71.6	34.6	82.7	33.3	80.5	32.1	78.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 15.7	13.2	11.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	12.2	72.3	3.5	1.8	9.6	73.6	7.6	74.7	5.6	75.7	3.1	77.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	34.7	37.0	43.1	50.0	58.0	62.9	66.1	65.6	59.7	52.1	42.0	35.6
	DBStd	13.24	7.61	7.26	7.43	7.21	6.67	6.60	6.06	6.10	6.12	6.63	6.53	7.05
	HDD50	1935	476	364	235	89	11	1	0	0	4	59	249	447
	HDD65	5516	940	783	679	452	233	118	62	67	178	402	691	911
	CDD50	2165	1	1	21	87	260	388	498	483	295	122	8	1
	CDD65	274	0	0	0	1	17	55	96	85	20	0	0	0
	CDH74	2352	0	0	0	23	174	472	819	742	119	3	0	0
	CDH80	636	0	0	0	0	31	101	237	261	6	0	0	0
(14) Wind	WSAvg	4.4	5.0	5.3	5.7	4.8	4.2	4.0	4.2	3.8	3.5	3.7	4.1	4.9
(15) Precipitation	PrecAvg	57.9	5.5	4.9	4.5	3.8	4.8	4.2	4.3	4.5	4.3	5.2	5.6	6.4
	PrecMax	75.6	12.1	10.7	14.8	7.6	8.2	8.8	9.1	10.9	8.7	9.7	13.5	13.5
	PrecMin	41.8	1.2	0.8	0.9	1.0	1.7	1.9	1.4	1.0	1.3	0.6	1.1	0.4
	PrecStd	8.7	2.9	2.7	3.2	2.0	1.7	2.0	1.8	2.2	2.0	2.4	3.0	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	58.7	68.1	77.4	83.6	86.7	90.6	95.8	81.2	72.8	61.4	54.6
		MCWB	48.5	48.1	51.8	58.9	67.5	68.1	69.0	67.1	66.0	62.0	54.4	49.7
	2%	DB	51.0	54.6	62.7	71.9	78.7	83.1	85.8	86.8	77.9	68.0	56.4	50.9
		MCWB	47.9	46.2	50.6	56.4	64.1	67.4	67.9	66.8	65.2	59.5	51.5	47.9
	5%	DB	48.6	51.0	58.3	67.1	74.8	79.8	82.5	82.2	74.1	64.9	53.2	48.5
		MCWB	45.6	45.1	48.5	54.2	61.1	66.3	66.9	66.3	63.6	58.1	49.4	45.8
	10%	DB	45.6	47.9	54.2	63.0	70.8	76.1	79.1	77.6	70.5	61.5	50.6	45.9
		MCWB	43.4	43.7	47.2	52.3	59.1	64.3	66.1	65.3	61.7	56.3	47.6	44.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	49.3	54.6	60.1	68.6	71.1	71.3	70.7	68.9	63.5	54.9	51.2
		MCDB	52.0	56.0	63.4	73.4	81.9	82.2	84.1	86.5	77.2	70.0	59.5	52.9
	2%	WB	48.3	47.5	51.9	57.4	65.6	69.2	69.7	68.8	66.7	60.8	52.2	48.4
		MCDB	50.8	52.5	58.5	69.6	76.7	80.2	82.5	82.9	75.3	65.8	55.4	50.5
	5%	WB	45.9	46.0	50.1	55.3	63.1	67.4	68.3	67.2	64.6	58.6	50.0	46.2
		MCDB	48.0	50.5	55.5	65.2	71.9	77.7	80.6	79.4	72.5	63.6	52.6	48.2
	10%	WB	43.5	43.9	48.3	53.2	60.6	65.4	66.5	65.7	62.4	56.7	48.1	44.3
		MCDB	45.5	47.7	54.1	61.6	68.0	74.0	76.9	76.0	68.9	61.1	50.3	45.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	11.8	14.6	17.4	17.7	18.5	18.2	17.7	16.6	14.0	10.4	8.3
		MCDBR	11.2	16.7	21.6	25.2	24.3	23.6	24.2	25.3	21.6	19.0	14.4	11.1
		MCWBR	8.4	9.9	11.8	12.6	12.0	11.2	10.4	10.4	11.2	11.0	9.7	8.3
	5% WB	MCDBR	9.8	14.2	17.5	23.3	21.6	22.0	22.1	22.0	19.2	17.5	13.7	10.1
		MCWBR	8.0	9.1	10.4	11.8	11.6	10.9	10.2	10.2	10.6	10.6	9.7	8.1
(40) Clear-Sky Solar Irradiance	taub		0.306	0.330	0.361	0.397	0.404	0.409	0.407	0.401	0.378	0.359	0.325	0.303
	taud		2.448	2.383	2.327	2.243	2.266	2.288	2.290	2.321	2.365	2.405	2.450	2.450
	Ebn at Noon		246	263	271	271	273	272	270	267	262	248	235	233
	Edh at Noon		21	28	34	41	41	41	40	37	32	27	21	19
(44) All-Sky Solar Radiation	RadAvg		334	589	981	1375	1603	1888	1795	1559	1181	734	385	269
	RadStd		46	96	153	205	162	172	163	149	115	66	46	53

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (96 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.68	+0.52	N/A	N/A	N/A	N/A

Nomenclature: See separate page